

Unit 5

Probability, Null Hypotheses, and p-values

Lecture notes for IB DP Biology and Chemistry

Course purpose	To help students move from raw experimental data to cautious, justified scientific claims.
Unit focus	Probability language, null models, p-values, significance levels, Type I/II errors, power, and common interpretation errors.
Central habit	Read a statistical result as a conditional statement about data under a model, not as direct proof that a hypothesis is true or false.

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1 Purpose of Unit 5

Units 1–4 built the foundations: measurement, descriptive statistics, distributions, and experimental design. Unit 5 introduces the logic that sits behind many statistical tests:

Assume a null model. Ask how unusual the observed data would be under that model.
Use that answer to judge how much tension there is between the data and the model.

This unit is not about memorising a list of tests. It is about understanding what a statistical test is trying to do.

Core thesis

A p-value is not the probability that the null hypothesis is true. It is the probability of observing data at least this extreme, assuming the null hypothesis and model assumptions are true.

By the end of this unit, students should be able to:

1. use probability language correctly, especially conditional probability;
2. distinguish scientific hypotheses from statistical hypotheses;
3. formulate null and alternative hypotheses for common biology and chemistry investigations;
4. explain what a test statistic is;
5. explain the meaning of a p-value without inverting the condition;
6. distinguish one-tailed and two-tailed tests;
7. explain significance level, Type I error, Type II error, and power;
8. avoid common false conclusions such as “ $p > 0.05$ means no effect”;
9. write cautious IB-style statistical conclusions.

2 Probability as model-based language

In everyday language, probability often means a vague feeling of confidence: “this is likely”, “this is unlikely”, or “I am almost sure”. In statistics, probability is more disciplined. It is attached to a **model**.

Biology example

If a seed has a germination probability of $p = 0.70$, this does not say what one particular seed must do. It says how the outcomes should behave across many similar seeds under the model.

For this course, probability should be read as:

Given a model, how often should this type of outcome occur?

2.1 Events and probabilities

An **event** is something that may or may not happen.

Situation	Event	Possible probability statement
Seed germination	A seed germinates	$P(\text{germination}) = 0.70$
Bacterial survival	A bacterium survives treatment	$P(\text{survival}) = 0.40$
Mendelian inheritance	Offspring shows recessive phenotype	$P(\text{recessive}) = 0.25$
Coin model	The coin lands heads	$P(\text{heads}) = 0.50$

A probability is not a guarantee. A probability describes expected behaviour under repeated comparable conditions.

2.2 Conditional probability

Many statistical mistakes come from ignoring what is being conditioned on.

$$P(A \mid B)$$

means:

The probability of A , **given that** B is true.

The order matters:

$$P(A \mid B) \neq P(B \mid A)$$

in general.

Rain analogy

Let R mean “it rained” and W mean “the ground is wet”.

$$P(W \mid R)$$

means: if it rained, how likely is wet ground?

$$P(R \mid W)$$

means: if the ground is wet, how likely is rain?

These are different questions. Wet ground may be caused by rain, sprinklers, cleaning, a leaking pipe, or irrigation.

This distinction will become essential when interpreting p-values.

3 Scientific hypotheses and statistical hypotheses

A scientific hypothesis is usually about a real process. A statistical hypothesis is a simplified mathematical statement used for analysis.

Investigation	Scientific hypothesis	Possible statistical hypothesis
Fertilizer and plant growth	Fertilizer improves plant growth.	The mean height is the same in fertilizer and control groups: $\mu_F = \mu_C$.
Temperature and enzyme activity	Temperature changes catalase activity.	Mean reaction rate differs among temperature groups.
Mendelian inheritance	The trait follows monohybrid inheritance.	Offspring phenotype counts follow a 3 : 1 ratio.
Antibiotic survival	Antibiotic reduces bacterial survival.	Survival status is independent of treatment group.

The statistical hypothesis is usually narrower than the scientific idea. It must be narrow because the data and test require a precise statement.

4 Null and alternative hypotheses

The **null hypothesis**, written H_0 , is the reference model used for comparison. It often represents “no difference”, “no association”, or “the expected distribution holds”.

The **alternative hypothesis**, written H_A , represents the kind of departure from the null model that the test is designed to detect.

Null hypothesis

The null hypothesis is not necessarily what the researcher believes. It is the model we temporarily assume in order to ask whether the observed data are surprising.

4.1 Common null hypotheses in IB science

Question type	Typical H_0	Typical H_A
Two means	The two population means are equal.	The two population means are different, or one is greater than the other.
Several means	All group means are equal.	At least one group mean differs.
Correlation/regression	The true slope or correlation is zero.	The true slope or correlation is not zero, or has a predicted direction.
Goodness of fit	Observed counts follow the expected ratio.	Observed counts do not follow the expected ratio.
Independence	Two categorical variables are independent.	The variables are associated.

4.2 Example: plant height

Research question:

Does fertilizer increase mean height of wheat seedlings after 14 days?

A one-sided formulation could be:

$$H_0 : \mu_F \leq \mu_C$$

$$H_A : \mu_F > \mu_C$$

A two-sided formulation could be:

$$H_0 : \mu_F = \mu_C$$

$$H_A : \mu_F \neq \mu_C$$

The choice depends on the scientific question before seeing the data. It should not be chosen after observing which direction looks convenient.

Do not choose the direction after seeing the result

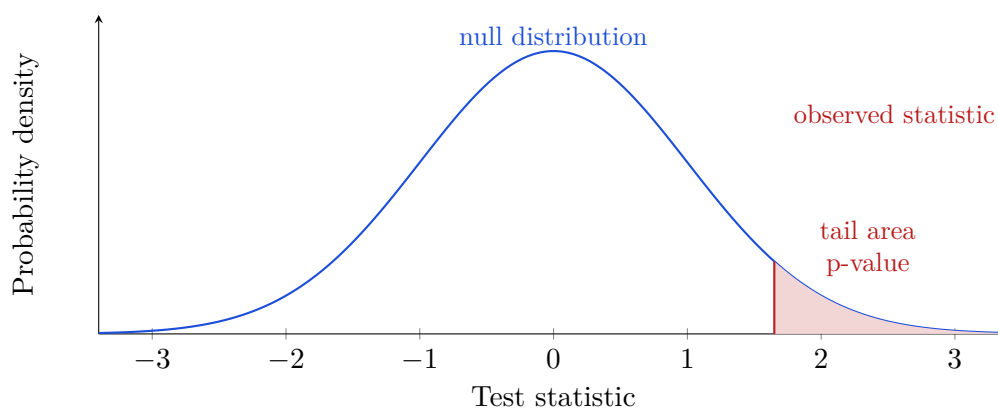
If the research question originally asked whether fertilizer changes growth, a two-tailed test is appropriate. Changing to a one-tailed test only because the observed result went upward is a form of post-hoc statistical manipulation.

5 Test statistics and null distributions

A **test statistic** is a number calculated from the data that measures departure from the null model. Examples include:

Test	Test statistic	What it measures
Two-sample t-test	t	Difference between means relative to sampling uncertainty.
Chi-square goodness-of-fit	χ^2	Deviation between observed and expected counts.
ANOVA	F	Between-group variation relative to within-group variation.
Regression	t for slope, or F for model	Whether the estimated relationship is large relative to uncertainty.

Under H_0 , the test statistic has an expected distribution. This is the **null distribution**. The p-value is found by locating the observed test statistic within this null distribution.

**Central idea**

A result is statistically surprising when the observed test statistic falls far into the tail of the null distribution.

6 What a p-value is

A p-value is a conditional probability:

$$\text{p-value} = P(\text{data at least this extreme} \mid H_0 \text{ is true and assumptions hold})$$

More precisely, it is the probability of obtaining a test statistic as extreme as, or more extreme than, the observed one, assuming the null hypothesis and the model assumptions are true.

What a p-value is not

A p-value is not:

- the probability that H_0 is true;
- the probability that H_A is true;
- the probability that the result happened “by chance”;
- the probability that the researcher made a mistake;
- a direct measure of effect size;
- a guarantee that the result is biologically or chemically important.

6.1 Correct and incorrect readings

Suppose a test gives:

$$p = 0.03$$

Correct reading:

If H_0 were true and the assumptions of the test were satisfied, data this extreme would occur about 3% of the time.

Incorrect reading:

There is a 3% chance that H_0 is true.

The incorrect reading reverses the condition.

$$P(\text{data} \mid H_0) \neq P(H_0 \mid \text{data})$$

7 Example: fair coin model

Suppose the null hypothesis is:

H_0 : the coin is fair

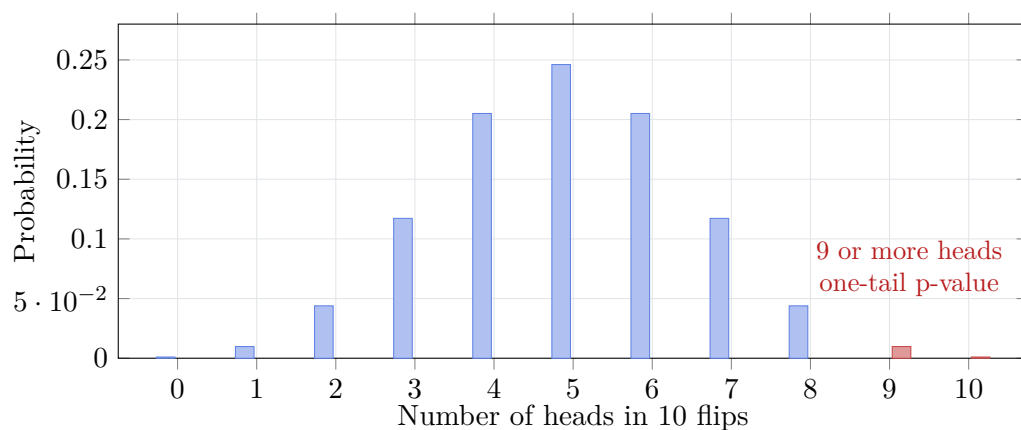
You flip the coin 10 times and observe 9 heads.

Under the fair-coin model, the number of heads follows:

$$X \sim \text{Binomial}(n = 10, p = 0.5)$$

The probability of 9 or more heads is:

$$P(X \geq 9) = P(X = 9) + P(X = 10) = \frac{10 + 1}{1024} = 0.0107$$



This small p-value says:

If the coin were fair, 9 or more heads in 10 flips would be rare.

It does not say:

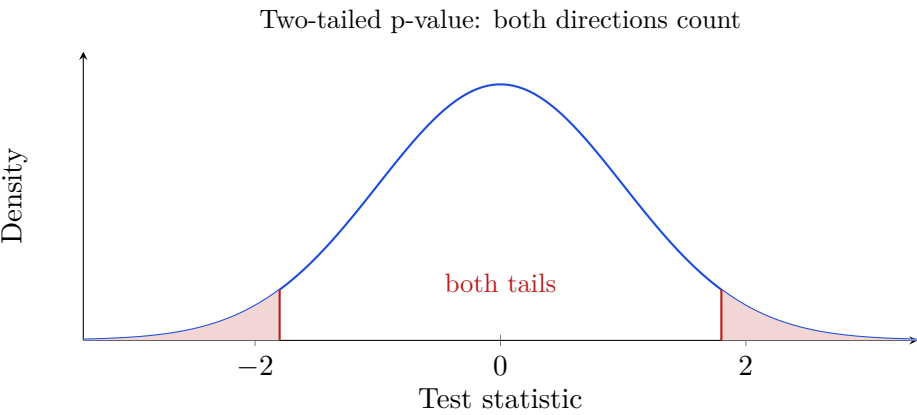
There is only a 1.07% chance that the coin is fair.

To estimate $P(H_0 \mid \text{data})$, one would need additional assumptions, such as how likely biased coins were before the experiment.

8 One-tailed and two-tailed tests

A **one-tailed test** looks for departure in a specified direction. A **two-tailed test** looks for departure in either direction.

Test type	Question	Example
One-tailed	Is the effect specifically greater than zero, or specifically less than zero?	Does fertilizer increase mean plant height?
Two-tailed	Is there any difference, regardless of direction?	Does fertilizer change mean plant height?



Direction must be chosen before the data are inspected

A one-tailed test is not a way to make a result look more significant. It is appropriate only when the research question genuinely predicts one direction and the opposite direction would not support the claim.

9 Significance level and decision rule

The **significance level**, written α , is the threshold used for deciding whether to reject H_0 . A common value is:

$\alpha = 0.05$

A common decision rule is:

if $p < \alpha$, reject H_0

if $p \geq \alpha$, fail to reject H_0

Important wording

Use “fail to reject H_0 ”, not “accept H_0 ”. A non-significant result does not prove that the null hypothesis is true.

9.1 Why 0.05 is not magic

The threshold 0.05 is a convention. It is not a law of nature. A result with $p = 0.049$ is not fundamentally different from a result with $p = 0.051$. Treating the boundary as a cliff encourages poor scientific reasoning.

p-value	Better interpretation
$p = 0.001$	Strong tension with H_0 , assuming the model is appropriate.
$p = 0.03$	Evidence against H_0 , but still interpret with effect size and design quality.
$p = 0.06$	Weak-to-borderline evidence; do not pretend it is either proof or nothing.
$p = 0.40$	Data are not unusual under H_0 ; this does not prove no effect.

10 Type I error, Type II error, and power

When a statistical test leads to a decision, two kinds of error are possible.

	H_0 is true	H_0 is false
Reject H_0	Type I error: false positive	Correct detection
Fail to reject H_0	Correct non-rejection	Type II error: false negative

10.1 Type I error

A **Type I error** occurs when we reject H_0 even though H_0 is true. If $\alpha = 0.05$, then the testing procedure is designed to have a 5% false-positive rate under the null model, assuming the assumptions hold and there is no p-hacking.

10.2 Type II error

A **Type II error** occurs when we fail to reject H_0 even though a real effect exists. This is common when sample size is small, variation is large, or the effect is weak.

10.3 Power

Power is the probability of detecting a real effect of a specified size:

$$\text{power} = 1 - \beta$$

where β is the probability of a Type II error.

Classroom interpretation

A small experiment with $n = 5$ plants per group may fail to detect a real fertilizer effect. The correct conclusion is not “fertilizer has no effect”. It is “this experiment did not detect an effect, and the small sample size gives low power.”

11 The p-value inversion problem

The correct p-value question is:

$$P(\text{data this extreme} \mid H_0)$$

The common wrong question is:

$$P(H_0 \mid \text{data})$$

These are not the same.

Modus tollens and statistics

In formal logic:

$$R \rightarrow W, \quad \neg W, \quad \therefore \neg R$$

If rain always makes the ground wet, then dry ground rules out rain.

In statistics the relationship is weaker:

$$H_0 \rightarrow \text{extreme data are unlikely}$$

Extreme data can still occur even when H_0 is true. Therefore a small p-value creates evidence against H_0 , but it does not logically prove H_0 false.

11.1 Bayesian comparison

To calculate the probability that H_0 is true after observing data, one would need more than the p-value. In Bayesian notation:

$$P(H_0 \mid D) = \frac{P(D \mid H_0)P(H_0)}{P(D)}$$

The p-value only resembles part of this expression. It does not provide the prior probability $P(H_0)$, and it does not automatically compare all alternative explanations.

12 Non-significant results

A common mistake is to treat $p > 0.05$ as proof of no effect.

Bad wording:

There was no difference between the groups.

Better wording:

No statistically significant difference was detected between the groups.

Best wording, if possible:

No statistically significant difference was detected. The observed mean difference was 2.1 cm, but the confidence interval was wide, so the experiment had limited ability to rule out a biologically meaningful effect.

Absence of evidence is not evidence of absence

A non-significant result may mean there is no effect. It may also mean the sample size was too small, the measurement was too noisy, the design was weak, or the chosen test was inappropriate.

13 Effect size must accompany p-values

A p-value does not tell us how large or important an effect is. Large samples can make tiny differences statistically significant. Small samples can fail to detect large but uncertain differences.

Result	Possible problem	What else to report
$p < 0.001$	Effect may be statistically clear but practically tiny.	Effect size and confidence interval.
$p = 0.20$	Effect may exist but experiment may be underpowered.	Observed effect size and uncertainty.
High R^2	Model may fit observed points but be inappropriate outside range.	Residuals, slope, uncertainty, and range of validity.
Significant chi-square	Association may be statistically real but weak.	Cramer's V , odds ratio, relative risk, or difference in proportions.

Reporting triad

For most IB science conclusions, train students to report:

effect size + uncertainty + statistical evidence

not just a p-value.

14 Multiple testing and p-hacking

If many tests are performed, some may become significant just by chance even when no real effects exist.

Suppose 100 independent null hypotheses are tested at:

$$\alpha = 0.05$$

If all null hypotheses are actually true, the expected number of false positives is:

$$100 \times 0.05 = 5$$

This is why multiple testing can generate impressive-looking but unreliable results.

14.1 Common p-hacking behaviours

- trying many tests and reporting only the significant one;
- removing outliers without a pre-defined rule;
- testing many subgroups until one gives $p < 0.05$;
- stopping data collection once a significant result appears;
- changing from a two-tailed to a one-tailed test after seeing the direction;
- changing the dependent variable after inspecting the data.

IB version of p-hacking

In school investigations, p-hacking often appears as post-hoc justification: students change the research question or statistical method after seeing which result looks best. This weakens the scientific argument.

15 Worked example: plant height

A student investigates whether fertilizer increases mean height of seedlings after 14 days.

Group	Heights / cm
Control	11.8, 12.1, 12.5, 12.6, 13.0
Fertilizer	13.9, 14.2, 15.2, 15.8, 16.4

Possible hypotheses:

$$H_0 : \mu_F \leq \mu_C$$

$$H_A : \mu_F > \mu_C$$

A software output might give:

$$p = 0.004$$

A weak conclusion would be:

The hypothesis is proven because $p < 0.05$.

A stronger conclusion would be:

The fertilizer group had a higher mean plant height than the control group. Under the null model of no increase, a difference this large would be unlikely. This supports the claim that fertilizer increased height under the tested conditions. However, the conclusion remains limited by the small sample size and the need to confirm that plant position, light exposure, and watering were controlled.

Notice that the stronger conclusion connects the p-value to the design and the biological claim.

16 Worked example: Mendelian ratio

A monohybrid cross is expected to produce a 3 : 1 ratio of dominant to recessive phenotypes.

Observed offspring:

Phenotype	Observed	Expected under 3 : 1
Dominant	68	75
Recessive	32	25

Hypotheses:

H_0 : offspring follow a 3 : 1 ratio

H_A : offspring do not follow a 3 : 1 ratio

The chi-square statistic is:

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

$$\chi^2 = \frac{(68 - 75)^2}{75} + \frac{(32 - 25)^2}{25} = 0.653 + 1.960 = 2.613$$

With $df = 1$, this gives approximately:

$$p \approx 0.106$$

Correct conclusion:

The data do not provide sufficient evidence to reject the 3 : 1 model at $\alpha = 0.05$. This does not prove the 3 : 1 ratio is true; it means the observed deviation is not unusual enough under the 3 : 1 model to reject it.

17 IB-style language templates

Weak wording	Better wording
The hypothesis was proven.	The data support the hypothesis under the tested conditions.
There was no effect.	No statistically significant effect was detected.
The result was due to chance.	The result would be unlikely under the null model.
The p-value shows the probability that the null is true.	The p-value shows how unusual the data would be if the null model were true.
The experiment was accurate because $p < 0.05$.	The result was statistically significant, but validity depends on design, measurement quality, and assumptions.
The graph proves a relationship.	The graph suggests a pattern; statistical analysis and uncertainty are needed to evaluate the claim.

Recommended conclusion structure

A strong conclusion often follows this order:

1. state the observed effect size or pattern;
2. state the statistical evidence;
3. translate it into the scientific claim;
4. state the uncertainty and main limitations;
5. avoid proof language.

18 Practice tasks

Task 1: Correct the interpretation

A student writes: “The p-value was 0.02, so there is a 2% chance that the null hypothesis is true.”

Rewrite this sentence correctly.

Task 2: Formulate hypotheses

For each investigation, write H_0 and H_A :

1. Does caffeine increase mean heart rate after 10 minutes?
2. Do observed pea plant phenotypes fit a 3 : 1 ratio?
3. Is bacterial survival associated with antibiotic treatment?
4. Does temperature change enzyme reaction rate?

Task 3: Choose one-tailed or two-tailed

For each question, decide whether a one-tailed or two-tailed test is more appropriate. Justify your answer.

1. Does fertilizer increase plant height?
2. Does pH affect enzyme activity?
3. Does a new method give different titration results from the old method?
4. Does a toxin reduce bacterial survival?

Task 4: Interpret non-significance

An experiment comparing two groups gives $p = 0.18$. The student writes: “Therefore, the treatment had no effect.”

Explain why this is too strong. Write a better conclusion.

19 Unit 5 checklist

Before moving to specific tests in later units, students should be able to answer these questions:

1. What is the null hypothesis?
2. What is the alternative hypothesis?
3. What kind of data were collected?
4. What test statistic is being used?
5. What distribution or model is assumed under H_0 ?
6. What does the p-value mean in this specific context?
7. Is the test one-tailed or two-tailed, and was that decided before seeing the data?
8. What is the effect size?
9. What uncertainty remains?
10. What conclusion is allowed, and what conclusion would overclaim?

Final message

A p-value is a tool for judging tension between data and a null model. It is not a truth meter. Good scientific reasoning requires p-values to be interpreted alongside effect size, uncertainty, assumptions, and experimental design.